

key

Can Touch This!



You just got hired to work at Bubbles Bottling Company which produces cans of Bubly sparkling water. You are working in the logistics department and have to make some decisions about potential new products.

1. A can of Bubly water has a height of 12.1 cm and a radius of 3.1 cm. Calculate its volume.
(Hint: Volume of a cylinder = $\pi r^2 h$)

$$V = \pi (3.1)^2 (12.1) = 365.31 \text{ cm}^3$$

2. Without changing the volume of the can, come up with two new can designs—a taller can and a wider can—and record their dimensions below.

Height	12.1 cm	13 cm	$\frac{365.3}{\pi (4.5)^2} = h$ $h = 5.74 \text{ cm}$
Radius	3.1 cm	$\frac{365.3}{13\pi} = r^2$ $r = 2.99 \text{ cm}$	4.5 cm
Volume	365.3 cm ³	365.3 cm ³	365.3 cm ³

$$h = \frac{V}{\pi r^2}$$

$$r = \sqrt{\frac{V}{\pi h}}$$

$$\begin{aligned} \pi (2r)^2 \left(\frac{1}{2}h\right) &= 4 \cdot \frac{1}{2} \pi r^2 h \\ &= 2\pi r^2 h \\ &= 2(\text{original}) \end{aligned}$$

A colleague argues that doubling the radius of the can and halving the height will keep the volume the same. Is this true? Why or why not?

No, this is not true. Because the radius gets squared it has a larger impact on volume than the height does. The height would need to be less than half.

4. Meijer sells Bubly water and is offering a President's Day sale on its individual cans. For the price of an original size can of Bubly you can choose either:
 - a. A can with the same height but twice the radius OR
 - b. A can with the same radius but twice the height

Which offer is the better deal? Explain.

Doubling the radius quadruples the volume!

Option a) is better because you get a can w/ 4x the volume of the original but in option b) you only double the volume.

5. Suppose you had the option of buying a can with twice the radius and twice the height for triple the price. Is this a better or worse deal than either of the two options above? Explain.

$\pi (2r)^2 (2h) = 8\pi r^2 h$
This option gives you 8x the volume but for triple the price. This is worse than option a) because 2 of those cans gives 8x original volume for double the price. It is better than option b) though.

Lesson 0.4 – Reasoning with Formulas

QuickNotes

Formulas can be rearranged to solve for a given variable.

Scaling dimensions does not always affect volume proportionally.

Common volume formulas:

Prisms/cylinders

$$V = (\text{Area of base})(\text{height})$$

cones/pyramids

$$V = \frac{1}{3}(\text{area of base})(\text{Height})$$

spheres

$$V = \frac{4}{3}\pi r^3$$

Check Your Understanding

1. If the volume of a sphere is $400\pi \text{ in}^3$, what is its radius? ($V = \frac{4}{3}\pi r^3$)

$$400\pi = \frac{4}{3}\pi r^3$$

$$300 = r^3$$

$$r = \sqrt[3]{300} = 6.69 \text{ in}$$

2. The surface area of a cylinder is given by $SA = 2\pi r^2 + 2\pi rh$. Write an equation for the height of the cylinder in terms of the surface area and radius.

$$SA = 2\pi r^2 + 2\pi rh$$

$$SA - 2\pi r^2 = 2\pi rh$$

$$\frac{SA - 2\pi r^2}{2\pi r} = h$$

3. The directions on a Funfetti cake mix suggest using either a 10-inch round pan or an 8-inch square pan. Which pan will give you a taller cake? Explain your reasoning.

Round

$$\begin{aligned} \text{Area of base} &= \pi \cdot 5^2 \\ &= \pi \cdot 25 \approx 78.5 \end{aligned}$$

Square

$$\text{Area of base} = 8^2 = 64$$

Since the volume is the same, the square pan will make taller cake because the base area is smaller.

4. (Multiple Choice) A supermarket sells their store brand coffee in two sizes of cylindrical cans. The first can has a volume of 140 cubic inches. The second can has a radius that is half as large as the first can, but twice the height. What is the volume of the second can? (The volume of a cylinder is given by $V = \pi r^2 h$.)

(A) 35 cubic inches

(B) 70 cubic inches

(C) 140 cubic inches

(D) 280 cubic inches

$$V_1 = \pi r^2 h = 140 \text{ in}^3$$

$$\begin{aligned} V_2 &= \pi \left(\frac{1}{2}r\right)^2 (2h) = \frac{1}{2}\pi r^2 h = \frac{1}{2}(140) \\ &= 70 \text{ in}^3 \end{aligned}$$